

RAASEPORI JUSSARO, Finland

WMO: 027570

Lat: 59.821N

Lon: 23.576E

Elev: 23

StdP: 101.05

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-12.4	-18.1	0.8	-15.0	-15.0	1.0	-10.8	18.8	0.7	17.9	1.1	6.4	50	0.821

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.3	23.7	20.0	22.1	18.9	20.7	18.0	20.9	23.2	19.8	21.8	18.5	20.3	5.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	14.7	22.5	19.0	13.8	21.4	17.7	12.7	19.9	60.4	23.2	56.8	21.9	52.5	20.4	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
17.2	15.5	14.0	DB	-16.6	25.2	3.8	2.3	-19.3	26.8	-21.5	28.1	-23.7	29.4	-26.4	31.0	
			WB	-16.0	20.9	4.1	1.5	-18.9	22.0	-21.3	22.9	-23.5	23.7	-26.5	24.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-3.3	-2.4	-0.7	3.4	9.1	13.4	17.4	16.8	13.4	7.5	4.1	0.8
	DBStd	7.84	4.61	4.90	3.18	2.29	3.23	2.36	2.84	2.23	2.55	3.06	3.44	4.10
	HDD10.0	1902	413	347	332	198	57	2	0	0	6	84	177	287
	HDD18.3	4291	671	581	590	448	288	147	53	57	149	334	427	545
	CDD10.0	690	0	0	0	0	27	105	230	212	107	9	0	0
	CDD18.3	37	0	0	0	0	0	1	25	11	0	0	0	0
	CDH23.3	56	0	0	0	0	0	1	48	7	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.9	7.7	7.4	7.0	5.9	5.6	5.9	5.5	6.2	6.8	7.6	8.4	9.0
Precipitation	PrecAvg	662	56	40	35	34	34	57	65	80	59	73	70	59
	PrecMax	977	117	102	66	71	73	101	149	160	125	129	140	114
	PrecMin	436	10	5	6	4	9	12	6	7	16	17	4	19
	PrecStd	117	25	22	16	18	16	24	36	36	31	34	31	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	4.7	6.6	11.6	19.6	21.1	26.0	24.2	19.2	14.0	10.1	7.9
		MCWB	4.3	2.4	3.6	7.7	12.5	15.4	20.7	20.9	16.6	12.4	9.5	7.1
	2%	DB	4.2	3.2	4.6	9.3	17.2	19.3	24.7	22.5	18.1	12.9	9.4	7.0
		MCWB	3.6	2.5	2.5	6.3	12.8	15.3	20.5	19.6	16.0	11.7	8.7	6.3
	5%	DB	3.4	2.7	3.4	7.8	15.5	18.1	23.3	21.2	17.2	12.1	8.6	6.1
		MCWB	2.9	2.1	2.1	5.3	11.8	14.8	19.9	18.7	15.5	11.0	7.9	5.3
	10%	DB	2.4	2.1	2.7	6.6	13.7	17.0	21.9	20.1	16.5	11.5	7.9	5.3
		MCWB	1.9	1.7	1.8	4.5	10.9	14.0	18.8	17.6	14.9	10.4	7.1	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	3.3	4.1	8.0	14.5	17.6	22.1	21.4	17.5	12.9	9.7	7.4
		MCDB	4.7	3.9	5.7	10.8	17.5	19.4	24.4	23.5	18.8	13.4	9.8	7.7
	2%	WB	3.8	2.8	3.1	6.7	13.3	16.5	21.4	20.4	16.6	12.2	9.0	6.4
		MCDB	4.1	3.1	4.0	8.9	16.6	18.4	23.9	22.1	17.7	12.7	9.3	6.8
	5%	WB	3.0	2.2	2.6	5.8	12.3	15.6	20.5	19.3	15.9	11.5	8.2	5.5
		MCDB	3.4	2.5	3.2	7.4	15.1	17.5	22.9	20.9	16.9	12.0	8.5	6.0
	10%	WB	2.0	1.7	2.0	4.9	11.1	14.6	19.5	18.1	15.2	10.6	7.3	4.7
		MCDB	2.4	2.0	2.5	6.2	13.4	16.4	21.6	19.6	16.2	11.4	7.8	5.2
Mean Daily Temperature Range	5% DB	MDBR	3.8	3.2	3.9	3.9	4.6	4.4	4.3	3.9	3.6	3.3	2.8	3.2
		MCDBR	3.0	2.6	4.1	5.8	6.8	6.0	5.6	4.7	4.3	3.3	2.9	3.1
		MCWBR	2.9	2.4	3.1	3.7	4.1	3.5	3.2	2.8	3.0	3.0	3.0	3.1
	5% WB	MCDBR	2.9	2.5	3.4	5.1	6.3	5.0	4.8	4.0	3.6	2.9	2.8	2.9
		MCWBR	2.9	2.4	2.7	3.5	4.1	3.3	3.1	2.9	2.8	3.0	3.0	3.0
Clear-Sky Solar Irradiance	taub	0.278	0.311	0.333	0.371	0.366	0.362	0.388	0.382	0.351	0.334	0.309	0.264	
	taud	2.154	2.215	2.281	2.319	2.413	2.467	2.417	2.426	2.497	2.409	2.222	2.090	
	Ebn at Noon	491	682	794	828	863	871	835	811	780	662	449	353	
	Edh at Noon	44	72	91	106	103	100	103	95	75	60	42	31	
All-Sky Solar Radiation	RadAvg	0.24	0.77	2.29	3.95	5.53	5.83	5.51	4.34	2.64	1.15	0.31	0.15	
	RadStd	0.04	0.14	0.27	0.41	0.54	0.38	0.47	0.43	0.32	0.16	0.06	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (58 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	-135	N/A	N/A	N/A

Nomenclature: See separate page